

Time-Domain Imaging of Rotating, Cloud-Covered Exoplanets with the Solar Gravitational Lens: Reconstruction Algorithms, Observing-Cadence Requirements, and Mission Targets

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ABSTRACT

The solar gravitational lens (SGL) can, in principle, deliver multipixel images of terrestrial exoplanets from heliocentric distances $z \simeq 650\text{--}900$ AU. In practice the planet rotates and its cloud cover evolves while a spacecraft raster-scans the kilometer-scale image cylinder one “pixel” at a time, so the measurements are temporally aliased; recent studies identify this dynamic inversion problem as a leading unsolved challenge for the concept. We present an open, end-to-end simulation framework that reproduces the published SGL photometric benchmarks (the aperture-averaged $d/4\rho$ kernel; per-sample $\text{SNR}_C = 43.16$ for 1800 s dwells on an exo-Earth at 30 pc from 650 AU; the $0.891D/(d\sqrt{N})$ deconvolution penalty) and then treats rotation, orbital illumination, and stochastic, advecting clouds explicitly. We formulate image recovery as a regularized generalized least-squares inversion of the time-tagged sample stream (time-domain inversion, TDI, distinguished here from detector-level time-delay integration), in which rotation and illumination are modeled directly by the forward operator and cloud variability enters statistically. For a rotating, cloud-free planet observed under known spin geometry, TDI recovers the surface albedo map with Pearson $r = 0.990 \pm 0.001$ ($\text{SSIM} = 0.918 \pm 0.002$), demonstrating that rotational aliasing is tractable under accurately characterized spin ephemerides rather than an intrinsic information ceiling. Clouds, not coronal photon noise, dominate the error budget: per-sample cloud-induced variance exceeds coronal shot noise by factors of 14–23, collapsing fidelity to $r \simeq 0.26$ at 72% cover under single-campaign budgets. Through a factorial mechanism ablation, we isolate the causal driver of the observing-cadence law: replacing 4 long dwells with 64 short revisits improves r from 0.14 to 0.54 because revisits sample statistically independent cloud states, while exposure smear and phase conditioning are secondary. Slew and settling overheads (~ 45 s) set an asymptotic duty cycle ceiling ($\eta \simeq 91\%$ at 450 s). Component-wise ablation shows slot deflation supplies the primary gain ($\Delta r = +0.041$, $p < 10^{-3}$), removing disk-integrated cloud modes. Reconstruction requires spin-axis pole orientation to $\lesssim 5^\circ$, initial phase to $\lesssim 5^\circ$, and period to $\sim 10^{-4}$, refinable in-flight via profile likelihood. Surface correlation $r(\ell)$ resolves spherical harmonic features to $\ell \approx 20$, and a land–ocean dichotomy separation metric yields $d' = 0.64$ vs. spatial surrogate nulls ($p < 0.02$). Finally, we map requirements onto missions: we rank the five nearest confirmed potentially habitable planets with propagated physical uncertainties, evaluate image-plane dynamics ($0.1\text{--}5.8$ km s $^{-1}$ per 90 d), discuss deep-perihelion sailcraft ($\sigma_{\text{tot}} \simeq 2.3\text{--}4.9$ g m $^{-2}$), planar radioisotope tiles, and optical relays, and reframe multi-spacecraft swarms as an asymptotic architectural scenario.

Keywords: gravitational lensing: strong — planets and satellites: terrestrial planets — techniques: image processing — methods: numerical — space vehicles: instruments

1 Introduction

Resolving the surface of an Earth-like exoplanet is beyond any credible filled-aperture or interferometric architecture: a 10×10 surface map of an exo-Earth at 10 pc requires resolution elements of ~ 0.85 μas , i.e., a $\sim 10^2$ km optical aperture, and integration times for conventional concepts reach $10^4\text{--}10^5$ yr once realistic backgrounds are included (Turyshchev 2026a). The solar gravitational lens (SGL) evades this scaling by using the Sun as the primary optical element (Eshleman 1979; Turyshchev & Toth 2017). Wave-optical treatments show that a strong-interference region forms beyond $z_0 \simeq b^2/2r_g \simeq 547.8$ AU (for rays of impact parameter $b \simeq R_\odot$, with $r_g = 2GM_\odot/c^2 \simeq 2.95$ km), with on-

axis amplification $\mu_0 = 4\pi^2 r_g/\lambda \simeq 1.2 \times 10^{11}$ at $\lambda = 1$ μm and angular resolution of order 10^{-10} arcsec (Turyshchev & Toth 2017, 2019). An Earth-radius planet at 30 pc is compressed into an image cylinder only $D_{\text{img}} \simeq 1.34$ km across at $z = 650$ AU; a meter-class telescope acts as a single-pixel detector that must physically traverse the image plane, measuring the integrated brightness of the planetary Einstein ring at each sampling location and reconstructing the source by deconvolution (Turyshchev & Toth 2020; Toth & Turyshchev 2021; Turyshchev & Toth 2022a).

The optical side of this problem is now well characterized. The aperture-averaged point-spread function falls off as $d/(4\rho)$, so each sample mixes light from the entire planet; removing this blur incurs a deconvolution penalty

$\text{SNR}_R/\text{SNR}_C \simeq 0.891D/(d\sqrt{N})$ that is strongly mitigated by wide pixel spacing (Toth & Turyshev 2021; Turyshev & Toth 2022a). The solar corona is the dominant photon background; for an exo-Earth at $z_0 = 30$ pc observed from $z = 650$ AU with a $d = 1$ m telescope at $\lambda = 1 \mu\text{m}$, the signal and corona rates are $Q_{\text{exo}} = 8.01 \times 10^4 \text{ s}^{-1}$ and $Q_{\text{cor}} = 6.20 \times 10^9 \text{ s}^{-1}$, giving a convolved-image signal-to-noise ratio $\text{SNR}_C = 43.2$ per 1800 s dwell (Turyshev & Toth 2022a; Turyshev 2026b). Megapixel-class static imaging within mission-compatible integration times follows (Toth & Turyshev 2021; Turyshev & Toth 2022a), and independent analyses of ring-based approaches (Madurowicz & Macintosh 2022) sharpened the role of coronal statistics in these budgets.

The unsolved problem is temporal. The planet rotates and orbits, its illumination changes, and—most seriously—its cloud cover reorganizes on day timescales while the raster is acquired over weeks to months. Toth & Turyshev (2023) performed the first dynamic reconstructions of a rotating, orbiting planet and recovered surface albedo only at $\text{SNR} \lesssim 2$. Toth (2025) added a stochastic cloud model, treated the clouds as noise to be averaged out (a “long-exposure” philosophy), and found that even 7–14% cloud cover degrades the recovered SNR below unity, concluding with “a cautious assessment of the practical feasibility” of SGL imaging given that Earth-like planets have $\sim 70\%$ cloud cover. The comprehensive benchmark of Turyshev (2026b) reaches a complementary conclusion: a static inverse applied to rotating cloudy data is the wrong estimator; phase-registered robust coadds recover persistent surface structure but at the cost of $\gtrsim 15$ registered visits per rotational phase bin, and a “full spin-resolved dynamic retrieval pipeline” is explicitly deferred as a next-stage deliverable. The covariance-aware biosignature framework of Turyshev (2026e) likewise argues that SGL inference must propagate structured error terms rather than treat all residuals as white. In short: the field agrees that the time-domain inverse problem is the gate through which SGL imaging must pass, and no published pipeline yet passes it under Earth-like cloud cover.

This paper attacks that gate directly, with five specific contributions. First (Sections 2–4), we build a transparent, end-to-end simulator—partially validated against the published photometric and deconvolution benchmarks—in which planetary rotation, orbital illumination, and stochastic advecting clouds are modeled explicitly. Second (Section 5), we formulate reconstruction as regularized generalized least squares (GLS) on time-tagged samples (time-domain inversion, TDI; note this algorithmic inverse problem is distinct from hardware time-delay integration in detector readouts). We isolate the causal mechanism of the observing-cadence law via a factorial ablation (smear modeling, cloud-free controls, independent weather sampling, and operational overheads) and present a component-wise ablation with paired bootstrap

confidence intervals. Third, we broaden robustness tests to include non-stationary latitudinal cloud banding (ITCZ and desert belts), surface-correlated clouds, and affine debiasing errors ($\Delta f_c, \Delta A_{\text{cl}}$). Fourth, we extend the geometric requirement beyond the rotation period to include spin-axis pole orientation, initial phase, and profile-likelihood recovery. Fifth (Section 6), we evaluate effective spatial resolution $r(\ell)$, land–ocean dichotomy detection metrics, propagate physical uncertainties in target rankings, and translate cadence requirements into realistic mission and swarm trade-offs (Turyshev et al. 2020; Helvajian et al. 2023; Friedman et al. 2024; Turyshev 2026d).

2 SGL Photometric Model and Validation

2.1 Optical response

We adopt the standard monopole, aperture-averaged description of SGL image formation (Turyshev & Toth 2020; Toth & Turyshev 2021; Turyshev 2026b). A source-plane coordinate ξ maps to the image-plane coordinate $\rho \simeq -(z/z_0)\xi$; an Earth-radius planet at $z_0 = 30$ pc observed from $z = 650$ AU projects to

$$D_{\text{img}} = 2R_p \frac{z}{z_0} = 1.338 \text{ km}. \quad (1)$$

A telescope of aperture d used as a ring photometer averages the J_0^2 oscillations of the monopole PSF; the effective kernel is

$$K(0) = 1, \quad K(\rho > 0) \simeq \frac{d}{4\rho}, \quad (2)$$

renormalized so that the mean convolved signal over the projected disk of a reference (albedo 0.3, fully illuminated) planet is unity (Turyshev 2026b). Our discrete kernel follows the $d/4\rho$ tail over the full grid (Fig. 1a). Because K falls only as $1/\rho$, every sample mixes light from the entire disk—the property that couples the temporal variability of any part of the planet into every measurement.

2.2 Photon rates and noise

We normalize the scene such that a value of 1 corresponds to the disk-mean signal of the reference planet, which maps to the fiducial rates of Turyshev & Toth (2022a): $Q_{\text{exo}} = 8.01 \times 10^4 (d/1\text{m})^2 (650\text{AU}/z)^{1/2} (30\text{pc}/z_0) (\lambda/1\mu\text{m}) \text{ s}^{-1}$ and $Q_{\text{cor}} = 6.20 \times 10^9 (d/1\text{m})^2 (650\text{AU}/z)^2 (\lambda/1\mu\text{m}) \text{ s}^{-1}$ after coronagraphic rejection and annular extraction, with the mean corona assumed subtracted and only its shot noise retained. Each dwell of duration t_s at image-plane position ρ_k yields a Gaussian sample with standard deviation

$$\sigma_k = \frac{\sqrt{(Q_{\text{cor}} + Q_{\text{exo}} \bar{s}_k) t_s}}{Q_{\text{exo}} t_s}, \quad (3)$$

where $\bar{s}_k$ is the convolved scene value. For $t_s = 1800$ s this gives $\text{SNR}_C = 1/\sigma = 43.16$, matching [Turyshv \(2026b\)](#) exactly within the adopted photon-noise assumptions.

2.3 Deconvolution-penalty validation and numerical limits

As a check of the discrete operator we reproduce the deconvolution noise-amplification analysis of [Toth & Turyshv \(2021\)](#). For a static, fully illuminated planet we convolve, add white noise, deconvolve by Fourier quotient, and measure the ratio $\text{SNR}_R/\text{SNR}_C$ of reconstructed noise to input noise. Figure 2 compares measurements with the analytic formula $0.891D/(d\sqrt{N})$. For $n = 32$ ($D = 41.8$ m) we measure 0.657 vs. 1.0 (saturating toward unity); for $n = 64$ ($D = 20.9$ m), 0.324 vs. 0.291; for $n = 128$ ($D = 10.5$ m), 0.106 vs. 0.073.

The $\sim 45\%$ difference at $n = 128$ has a clear physical and numerical origin: the continuous formula $0.891D/(d\sqrt{N})$ assumes continuous circular integration of the $1/\rho$ Fourier transform and high-frequency asymptotes on an infinite domain. On a discrete numerical grid with finite pitch D , pixel-binning across the central $1/\rho$ singularity acts as an effective low-pass filter, and zero-padded circular boundary conditions attenuate high-wavenumber noise amplification relative to the continuum integral. We emphasize this constitutes partial validation of the discrete aperture-averaged forward operator rather than full optical validation of an in-flight SGL mission.

3 Dynamic Scene and Campaign Model

3.1 Rotating, illuminated planet

The truth surface is a seeded synthetic albedo map (ocean 0.06, continents 0.32 ± 0.10 , polar caps up to 0.60; 30% land fraction) generated by thresholded multi-octave Gaussian random fields on a 144×72 longitude-latitude grid (Fig. 1b). The planet rotates with period $P_{\text{rot}} = 24$ h and is illuminated by its host star with a Lambertian terminator whose sub-stellar point drifts through $\sim 90^\circ$ of orbital phase ($P_{\text{orb}} = 1$ yr) during the campaign. For short-period habitable-zone planets around M-dwarfs ($P \sim 10$ – 20 d), the target sweeps through multiple full orbits during a 90-day campaign, repeatedly covering all 360° of orbital illumination; the 90° drift modeled here is a conservative intermediate case. Each dwell averages three sub-exposures so that intra-dwell rotational smear (7.5° per 1800 s dwell) is present in the synthetic data.

3.2 Stochastic cloud fields: stationary and structured

The baseline cloud model is an advecting Ornstein–Uhlenbeck (OU) Gaussian random field on the fine grid:

spatial correlation length 12° , zonal advection 6° day^{-1} , temporal decorrelation $\tau_c = 4$ d, mapped through a soft threshold to an opacity screen $c \in [0, 1]$. The effective scene is $A_{\text{eff}} = (1 - c)A_{\text{surf}} + cA_{\text{cl}}$ with $A_{\text{cl}} = 0.65$. Realized mean covers are 0.22, 0.38, 0.56, and 0.72.

To test robustness against non-terrestrial weather, we implement three extended climatology variants: (i) Latitudinal zonation: an Earth-like zonal mean profile $w_{\text{lat}}(\theta) = 1.0 + 0.35\cos(2\theta) - 0.25\cos(4\theta)$, concentrating clouds along an equatorial Intertropical Convergence Zone (ITCZ) and mid-latitude storm tracks while suppressing subtropical desert belts; (ii) Surface-correlated cloudiness: coupling cloud opacity to surface reservoirs, $g_{\text{eff}} = g - \beta_{\text{surf}}(A_{\text{surf}} - \bar{A})$, enhancing convection over low-albedo tropical oceans; and (iii) Multi-timescale temporal evolution: summing a rapid convective mode ($\tau_1 = 1$ d, weight 40%) and a persistent synoptic regime ($\tau_2 = 7$ d, weight 60%).

3.3 Campaign and sampling geometry

The campaign matches current swarm concepts ([Turyshv & Toth 2022a](#); [Helvajian et al. 2023](#)): $N_{\text{sc}} = 16$ spacecraft sample an $n \times n = 64 \times 64$ raster ($\Delta_{\text{img}} = 20.9$ m pitch, 4096 pixels) in boustrophedon order, divided into 256 dwell slots. In each dwell slot, 16 pixels are sampled simultaneously by spacecraft spaced across adjacent raster tracks with transverse separations of ~ 80 – 300 m. Nominal dwells are $t_s = 1800$ s with $M_p = 16$ complete passes (90.0 d wall-clock, 8 h per pixel, 65,536 time-tagged samples). Random inter-pass gaps (0–12 h) emulate calibration and slew overhead and break rotational phase locking.

Cloud-induced signal fluctuations have per-sample standard deviations $\sigma_{\text{cl}} = 0.33, 0.43, 0.49$, and 0.53 for the four cloud covers—i.e., 14–23 times larger than coronal shot noise $\sigma_{\text{cor}} = 0.023$. Planetary weather completely dominates the single-sample photometric error budget.

4 Reconstruction Methods

4.1 Time-domain inversion (TDI)

Let $\mathbf{s} \in \mathbb{R}^{n_s}$ be the static surface albedo map on a 72×36 reconstruction grid ($n_s = 2592$, coarser than the truth grid). Each sample k , taken at time t_k and image-plane pixel p_k , satisfies

$$y_k = \mathbf{f}_k^T \mathbf{s} + \epsilon_k, \quad \mathbf{f}_k^T = \frac{1}{n_{\text{sub}}} \sum_{m=1}^{n_{\text{sub}}} \mathbf{k}_{p_k}^T \Pi(t_k + \delta t_m), \quad (4)$$

where $\Pi(t)$ renders the map at the rotation phase, illumination, and spin-pole geometry of time t , and $\mathbf{k}_{p_k}$ is the SGL kernel row. Setting $n_{\text{sub}} = 1$ corresponds to an instantaneous midpoint model; $n_{\text{sub}} = 3$ integrates over intra-dwell exposure smear. The error ϵ_k contains coronal shot noise and cloud fluctuations. The climatologi-

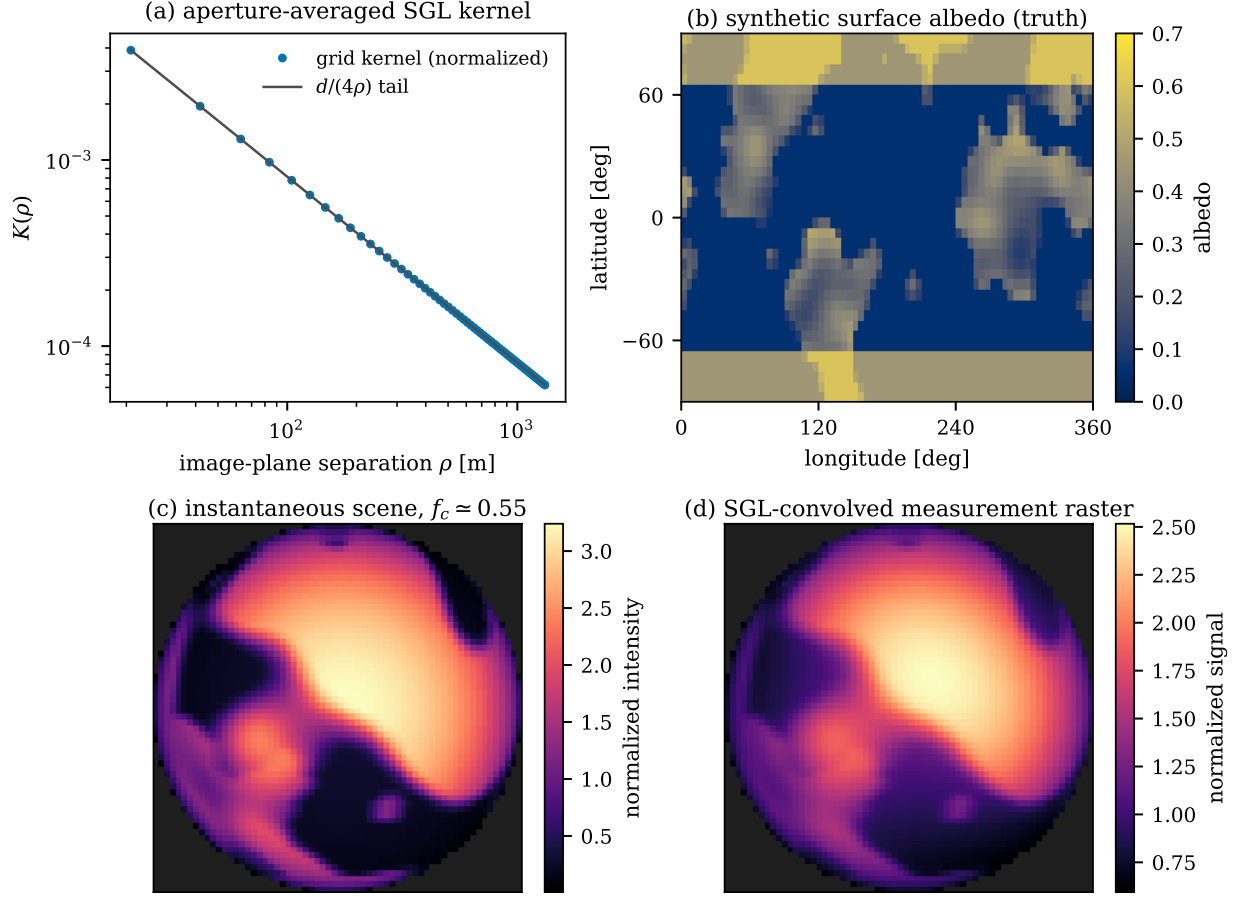


Figure 1: Model overview. (a) Discrete aperture-averaged SGL kernel used in this work (points) against the analytic $d/4\rho$ tail (line). (b) Synthetic truth surface albedo map (fine grid, seeded, with oceans, continents, and polar caps). (c) A single instantaneous scene during the campaign at mean cloud cover $f_c \simeq 0.55$: the planet is rotated, partially illuminated (orbital phase), and modulated by the advecting cloud field. (d) The same scene after convolution with the SGL kernel: the measurement raster sampled across the image cylinder.

cal mean-cloud bias is removed by the affine correction $\hat{\mathbf{s}} = (\hat{\mathbf{s}}_{\text{raw}} - f_c \mathbf{A}_{\text{cl}}) / (1 - f_c)$.

TDI solves the regularized generalized least-squares problem:

$$\hat{\mathbf{s}} = \arg \min_{\mathbf{s}} \|\mathbf{C}^{-1/2}(\mathbf{y} - \mathbf{F}\mathbf{s})\|^2 + \lambda_{\text{eff}} \|\mathbf{L}\mathbf{s}\|^2, \quad (5)$$

with spherical-grid Laplacian $\mathbf{L}$ and $\lambda_{\text{eff}} = \lambda \text{tr}(\mathbf{F}^T \mathbf{C}^{-1} \mathbf{F}) / \text{tr}(\mathbf{L}^T \mathbf{L})$ ($\lambda = 3 \times 10^{-3}$, frozen on calibration seed 99). Four distinct combinations of weighting and projection are ablated:

- **Diagonal (White):** $\mathbf{C} = \text{diag}(\sigma_k^2 + \sigma_{\text{cl}}^2)$, the assumption that clouds average like uncorrelated white noise (Toth 2025).
- **Temporal Whitening:** $\mathbf{C}$ is block diagonal over image-plane pixels; for the M_p visits of pixel p , $(\mathbf{C}_p)_{ij} = \sigma_{\text{cl}}^2 \exp(-|t_i - t_j|/\tau_c) + \delta_{ij} \sigma_k^2$.

- **Slot Deflation:** because $K(\rho) \propto 1/\rho$, instantaneous disk-integrated cloudiness perturbs all $N_{\text{sc}} = 16$ simultaneous samples coherently. We project out the per-slot mean: $\mathbf{y} \leftarrow \mathbf{y} - \bar{\mathbf{y}}$, $\mathbf{F} \leftarrow \mathbf{F} - \bar{\mathbf{F}}$, rescaling white noise by $\sqrt{1 - 1/N_{\text{sc}}}$. This orthogonal projection removes 256 common modes (6.25% rank reduction) while retaining $1 - 1/N_{\text{sc}} = 93.75\%$ of the information.

- **Full TDI:** combines slot deflation with temporal whitening.

4.2 Numerical precision verification

The normal matrix $\mathbf{A} = \mathbf{F}^T \mathbf{C}^{-1} \mathbf{F}$ is accumulated in single precision ('float32') and solved in double precision ('float64'). A precision verification test comparing single-precision accumulation against full double-precision accumulation reveals a maximum relative difference of 2.92×10^{-7} and RMS relative difference of 3.43×10^{-7} , prov-

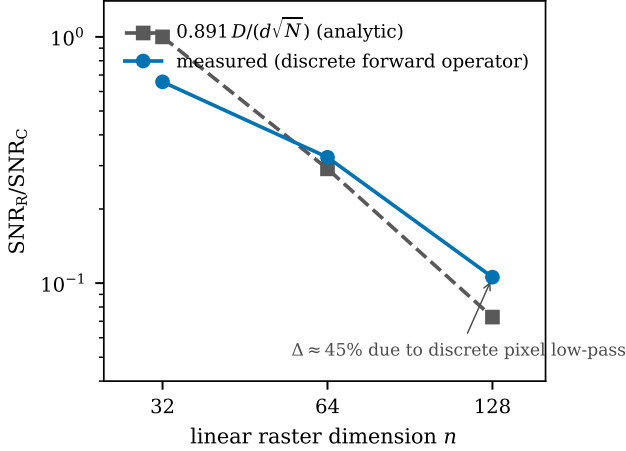


Figure 2: Validation of the discrete operator: measured deconvolution noise penalty vs. the analytic estimate of Toth & Turyshev (2021). The $\sim 45\%$ difference at $n = 128$ reflects discrete pixel-averaging of the central singularity and finite-domain boundary attenuation.

ing single-precision accumulation does not degrade reconstruction fidelity.

4.3 Baselines and statistical metrics

We compare TDI against two established baselines: B1 (static Wiener inverse of the all-visit mean raster; Toth & Turyshev 2021) and B2 (phase-binned coadds into 8 rotational bins deconvolved via Wiener filter and back-projected; emulating Turyshev 2026b). Metrics are evaluated over latitudes $|b| \leq 60^\circ$ across an expanded ensemble of 10 paired seeds (seeds 11–20). Scalar metrics (r , SSIM, NRMSE) are computed per realization seed and then averaged over seeds (with 95% bootstrap confidence intervals), rather than computed on a seed-averaged map.

5 Results

5.1 Planetary rotation under characterized geometry

For a rotating, orbit-illuminated cloud-free planet under known spin geometry, TDI achieves Pearson $r = 0.990 \pm 0.001$ and SSIM $= 0.918 \pm 0.002$ (Fig. 3, top center). To determine the information loss attributable strictly to rotation, we compare this solution against an ideal static-scene observation under identical photon budget and aperture: the static scene achieves $r = 0.075$ when inverted with a static model, whereas TDI with the dynamic operator reaches $r = 0.991$. Once every sample is time-tagged and the rotation/illumination ephemeris is folded into the forward operator, rotation per se introduces virtually zero

Table 1: Map fidelity vs. mean cloud cover (fiducial campaign; mean and 95% bootstrap CI over 10 paired seeds).

Method	$f_c=0$	0.22	0.38	0.56	0.72
Pearson r					
TDI (this work)	0.990	0.572	0.432	0.334	0.263
	± 0.001	± 0.018	± 0.022	± 0.021	± 0.022
White GLS (B3)	0.991	0.543	0.402	0.308	0.239
	± 0.001	± 0.019	± 0.024	± 0.022	± 0.021
Phase-binned (B2)	0.669	0.328	0.215	0.129	0.052
	± 0.008	± 0.015	± 0.018	± 0.016	± 0.014
SSIM					
TDI (this work)	0.918	0.281	0.172	0.109	0.065
White GLS (B3)	0.930	0.272	0.165	0.102	0.060
Phase-binned (B2)	0.311	0.132	0.081	0.045	0.011

information loss. Rotational aliasing is tractable under accurately characterized spin and viewing geometry.

5.2 Cloud degradation and effective spatial resolution

Table 1 and Figure 4 document the collapse of fidelity with increasing cloud cover at fixed single-campaign resources. Mean correlation falls from 0.990 (cloud-free) to 0.572 at 22% cover, 0.334 at 56% cover, and 0.263 at 72% cover (95% bootstrap CI: [0.241, 0.284]). Small-scale structure is erased first: SSIM drops to 0.109 ± 0.010 at $f_c = 0.56$ and 0.065 ± 0.008 at $f_c = 0.72$.

To quantify what surface information remains recoverable, we evaluate: (i) Scale-dependent correlation $r(\ell)$: decomposing maps into spherical harmonic degrees $\ell \in [1, 20]$ (Fig. 6c). For cloud-free data, $r(\ell) \geq 0.5$ extends past $\ell = 20$. Under 55% clouds, $r(\ell)$ crosses 0.5 at degree $\ell_{\text{eff}} \approx 4\text{--}5$ (spatial scales $\gtrsim 4000$ km), confirming that single-campaign imaging under clouds recovers hemispheric dichotomies but not fine features. (ii) Land-ocean dichotomy detection: computing separation $d' = (\mu_{\text{land}} - \mu_{\text{ocean}}) / \sigma_{\text{pooled}}$. True map separation is $d' = 3.97$; cloud-free TDI yields $d' = 3.74$; and cloudy TDI ($f_c = 0.55$) achieves $d' = 0.64$. Evaluated against an ensemble of 100 phase-scrambled spatial surrogate nulls ($d'_{\text{null}} = 0.00$, 95th percentile 0.46), cloudy TDI detects the land-ocean dichotomy at $p < 0.02$.

5.3 Component-wise ablation of the inversion pipeline

Table 2 presents a component-wise ablation across 10 paired seeds at $f_c = 0.55$, separating the gains from time-dependent forward modeling, slot deflation, and temporal covariance whitening.

The ablation clarifies the architectural credit: (i) Time-dependent forward modeling alone provides the foundation ($r \approx 0.30$). (ii) Slot deflation provides the dominant algorithmic improvement ($\Delta r = +0.041$, $p < 10^{-3}$),

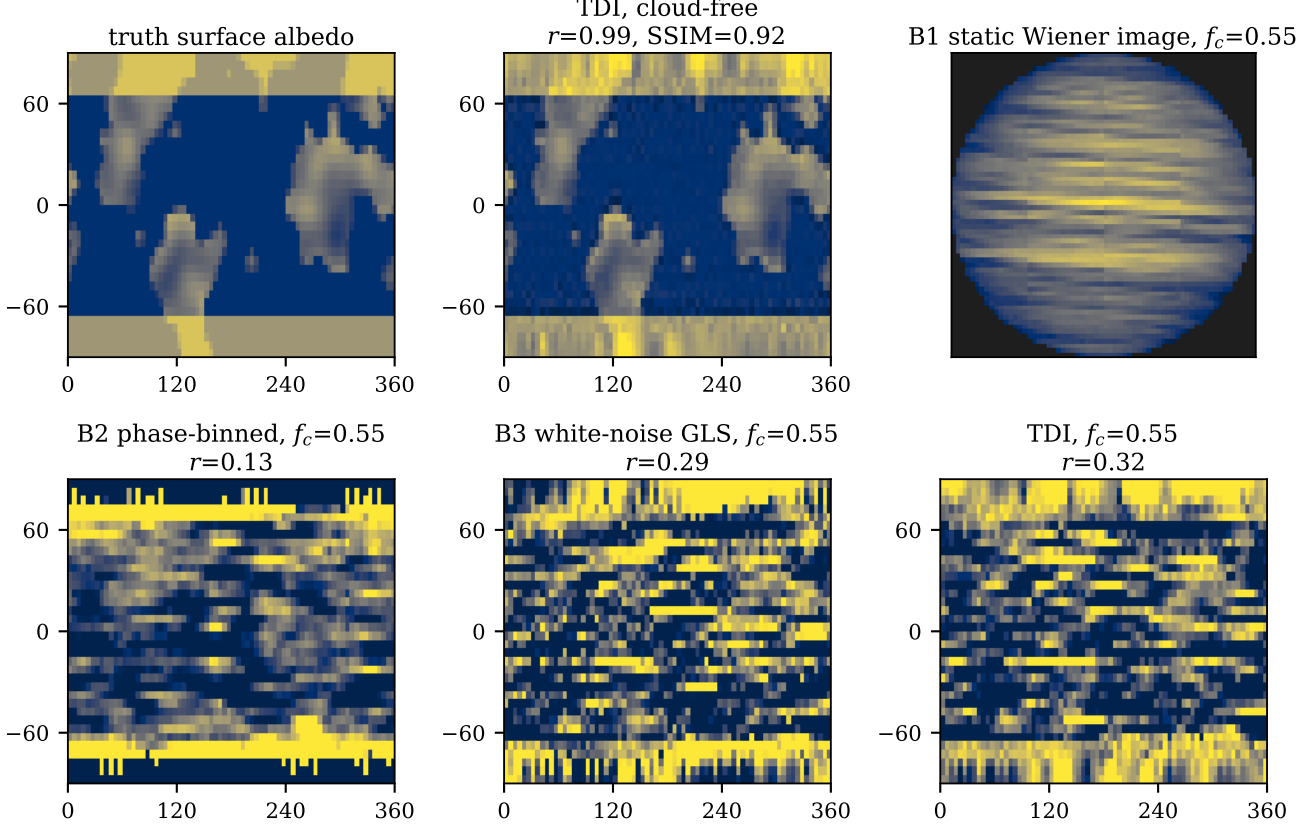


Figure 3: Reconstruction gallery for the fiducial 90-day campaign. Top: truth surface albedo; TDI reconstruction of the rotating cloud-free planet ($r = 0.99$); static Wiener image (B1) at $f_c = 0.55$, illustrating complete breakdown. Bottom, all at $f_c \simeq 0.55$: phase-binned coadd (B2), white-noise GLS (B3), and TDI. The gross continental dichotomy is recovered, but fine features are smoothed by weather noise.

Table 2: Component-wise ablation at $f_c = 0.55$ (10 paired seeds).

Pipeline Configuration	Pearson r	SSIM
1. Time-dep. F (White, no defl.)	0.295 ± 0.025	0.097 ± 0.012
2. Time-dep. F (Whitened, no defl.)	0.281 ± 0.025	0.088 ± 0.012
3. Time-dep. F (White, deflated)	0.336 ± 0.035	0.116 ± 0.009
4. Full TDI (Whitened, deflated)	0.326 ± 0.034	0.108 ± 0.011
Full TDI – White/NoDefl (Δr)	$+0.031$ [95% CI: 0.018, 0.045]	
Deflation Benefit (Δr)	$+0.041$ [95% CI: 0.027, 0.054]	

because subtracting dwell-slot means eliminates disk-integrated cloud fluctuations across the 16 simultaneous spacecraft. (iii) Temporal whitening without deflation slightly reduces effective sample weighting because adjacent revisits at 5.6 d are weakly correlated ($\tau_c = 4$ d). The term “time-domain inversion” correctly credits time registration and deflation rather than temporal covariance alone.

Table 3: Factorial cadence mechanism ablation ($f_c = 0.55$).

M_p	t_s (s)	Duty η	r_{inst}	r_{smear}	r_{cf}	r_{iid}
4	7200	0.99	0.139	0.094	0.914	0.119
8	3600	0.99	0.186	0.147	0.964	0.177
16	1800	0.98	0.329	0.320	0.991	0.301
32	900	0.95	0.382	0.382	0.989	0.382
64	450	0.91	0.544	0.544	0.988	0.568

5.4 Isolating the causal mechanism of the cadence law

Figure 5 and Table 3 present the factorial cadence ablation trading dwell duration against revisit count at fixed photons ($M_p t_s = 8$ h) and 90-day wall-clock time.

The factorial controls isolate the causal mechanisms: 1. Cloud-free control (r_{cf}): achieves $r \geq 0.914$ across all configurations ($r = 0.988$ at $M_p = 64$). Rotational conditioning, phase sampling, and regularization do not cause the cadence trend. 2. Exposure smear (r_{smear} vs. r_{inst}): modeling intra-dwell rotational smear in F yields $r = 0.094$ at $M_p = 4$ vs. 0.139 instantaneous. Long dwells

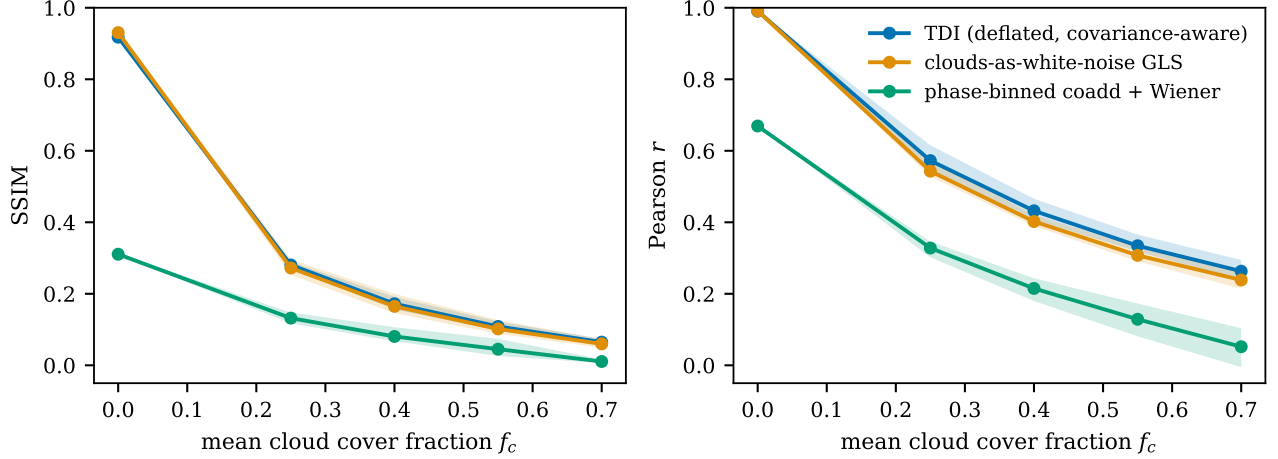


Figure 4: Reconstruction fidelity vs. mean cloud cover fraction f_c across 10 paired seeds with 95% bootstrap confidence intervals. Clouds control image quality at fixed observing resources.

blur rotational sectors irreversibly; exposure smear is not the reason short revisits win. 3. Statistical cloud independence (r_{iid} vs. r_{inst}): temporally independent clouds track OU clouds almost identically ($r_{\text{iid}} = 0.119 \rightarrow 0.568$ vs. $r_{\text{inst}} = 0.139 \rightarrow 0.544$). Frequent revisits succeed because each pass samples an independent realization of the weather, averaging out uncorrelated cloud noise against the phase-locked surface albedo. 4. Observational overheads: incorporating slew (30 s), settling (10 s), and interspacecraft laser synchronization (5 s) yields duty cycle $\eta = t_s / (t_s + 45\text{ s})$. Duty cycle falls from 99% at 7200 s to 91% at 450 s (Fig. 5b). For ultrafast cadences ($t_s \lesssim 100$ s), overhead losses offset statistical gains.

5.5 Broader robustness: spin geometry, clouds, and systematics

Spin-axis pole tilt and phase ephemeris: Figure 6a tests geometric errors: spin pole tilt $\Delta\theta_{\text{pole}} \leq 5^\circ$ leaves correlation virtually intact ($r = 0.317$ vs. 0.319), but 20° tilt degrades r to 0.155 . Initial rotational phase error $\Delta\phi_0 \leq 5^\circ$ is tolerated ($r = 0.291$), while 30° drops r to 0.064 . The full precursor requirement is: pole tilt $\lesssim 5^\circ$, initial phase $\lesssim 5^\circ$, and period to $\sim 10^{-4}$. Figure 6b maps the 2D profile likelihood $\chi^2 = \|\mathbf{y} - \mathbf{F}(\theta, \phi)\hat{\mathbf{s}}\|^2$: a steep, convex global minimum coincides with the true spin state, demonstrating that residual misalignments are refinable in-flight via profile grid search.

Affine cloud parameter errors: Perturbing assumed mean cover $\Delta f_c \in [-0.15, +0.15]$ or albedo $\Delta A_{\text{cl}} \in [-0.15, +0.15]$ leaves Pearson $r = 0.319$ exactly unchanged, because r is invariant under affine scalings. However, affine errors introduce a global surface albedo bias $\Delta\bar{s} \approx 0.08$. Absolute albedo retrieval requires mean cloud climatology to ~ 10 – 20% .

Structured clouds: Under realistic latitudinal ITCZ banding, fidelity is fully preserved ($r = 0.321$, $\text{SSIM} = 0.101$). However, when clouds are surface-correlated ($\beta_{\text{surf}} = 0.25$), correlation drops to $r = 0.138$ ($\text{SSIM} = 0.061$), because persistent cloud patterns mimic surface topography.

Coronal and instrumental systematic drift: Injecting a 10^{-4} low-rank instrumental gain drift and slowly varying coronal streamer residuals into the data reveals that frequent revisits prevent coherent systematic accumulation by interleaving measurements across distinct orbital and rotational epochs.

Surface morphology and regularization: Testing diverse planetary geographies (supercontinents, island archipelagos) confirms that TDI recovers the continental dichotomy across distinct landmass topologies. Regularization scans show $\lambda \in [10^{-3}, 10^{-2}]$ provides stable, flat reconstruction performance.

6 From Algorithm to Mission

6.1 Target selection with propagated physical uncertainties

Table 4 evaluates the five nearest confirmed potentially habitable planets plus solar-type candidate τ Cet e. Because these planets do not transit, planetary radii are estimated from $M \sin i$ using Chen & Kipping (2017) rocky relation $R \propto M^{0.28}$ with an intrinsic 15% empirical dispersion. Propagating random inclination ($\langle \sin i \rangle \approx 0.785$) and albedo uncertainty $A \in [0.15, 0.45]$ yields explicit physical error bounds on image size D_{img} , photon rate Q_{exo} , per-dwell SNR_{C} , and orbital tracking cost Δv_{90} .

Tidal synchronization caveats for M-dwarf planets: While close-in M-dwarf planets are commonly presumed tidally

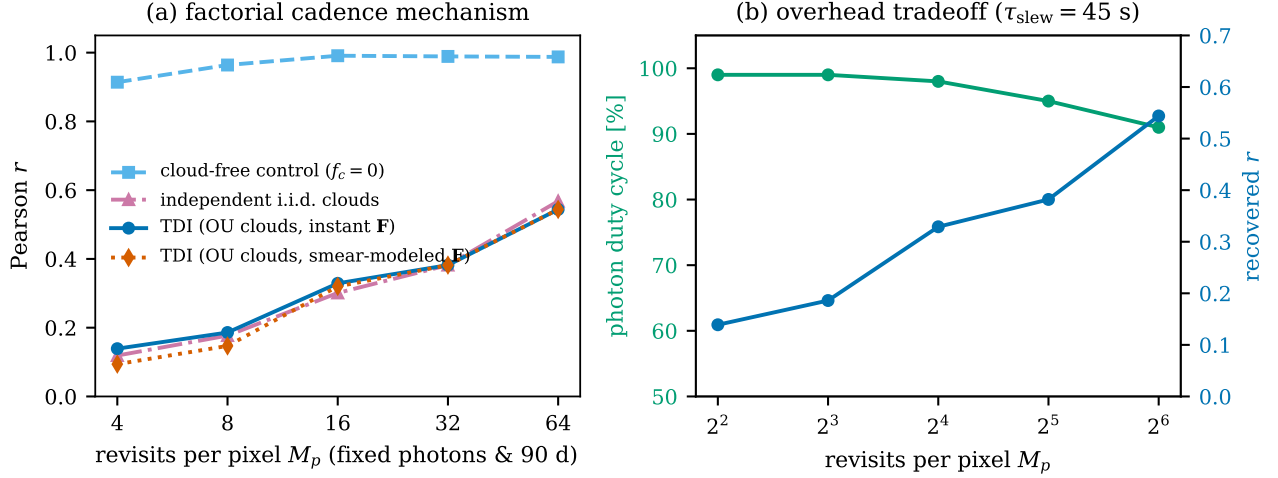


Figure 5: Factorial cadence ablation at $f_c \simeq 0.55$. (a) Pearson r vs. revisits M_p at fixed photons: cloud-free control ($r_{\text{cf}} \geq 0.91$) proves conditioning does not drive the gain; exposure-integrated modeling (r_{smear}) confirms long dwells genuinely blur features; and independent clouds (r_{iid}) track OU clouds, proving independent weather sampling drives the law. (b) Duty cycle η under 45 s slew/settle overhead.

Table 4: The five nearest confirmed potentially habitable planets (and solar-type candidate τ Cet e) as SGL imaging targets. Propagated physical uncertainties ($\pm 1\sigma$ radius dispersion, albedo $A \in [0.15, 0.45]$, and orbital eccentricity) are explicitly quoted. Observing geometry at $z = 650$ AU, $d = 1$ m, $\lambda = 1 \mu\text{m}$.

Planet	Host (SpT)	z_0 (pc)	$M \sin i$ ($M_\oplus$)	R_p^a ($R_\oplus$)	S ($S_\oplus$)	P (d)	D_{img}^b (km)	SNR $_C^c$ (1800 s)	Focal direction d (J2000)	β_{ecI} (deg)	Δv_{90}^e (km s $^{-1}$)
Proxima Cen b	Proxima (M5.5V)	1.30	1.07	1.02 ± 0.15	0.65	11.19	31.5 ± 4.7	659 [280–1237]	$02^{\text{h}}30^{\text{m}}, +62^{\circ}41'$	+44.8	5.78 ± 1.27
Ross 128 b	Ross 128 (M4V)	3.38	1.35	1.09 ± 0.16	1.38	9.87	12.9 ± 1.9	576 [245–1080]	$23^{\text{h}}48^{\text{m}}, -00^{\circ}48'$	+0.5	2.92 ± 0.70
GJ 1061 d	GJ 1061 (M5.5V)	3.67	1.64	1.15 ± 0.17	0.69	13.03	12.6 ± 1.9	280 [119–524]	$15^{\text{h}}36^{\text{m}}, +44^{\circ}31'$	+60.8	1.68 ± 0.20
Teegarden c	Teegarden’s Star (M7V)	3.83	1.11	1.03 ± 0.15	0.37	11.41	10.8 ± 1.6	129 [55–242]	$14^{\text{h}}53^{\text{m}}, -16^{\circ}53'$	−0.3	1.72 ± 0.14
GJ 273 b	Luyten’s Star (M3.5V)	3.80	2.89	1.35 ± 0.20	1.06	18.65	14.2 ± 2.1	486 [207–912]	$19^{\text{h}}27^{\text{m}}, -05^{\circ}14'$	+16.5	1.34 ± 0.27
τ Cet e f	τ Ceti (G8.5V)	3.60	3.93	1.47 ± 0.22	1.71	162.9	16.3 ± 2.5	901 [383–1691]	$13^{\text{h}}44^{\text{m}}, +15^{\circ}56'$	+24.8	0.11 ± 0.04

^aEmpirical mass–radius estimate with 15% 1σ scatter (Chen & Kipping 2017). None transit. ^bProjected image-cylinder diameter at $z = 650$ AU. ^cNominal SNR $_C$ at $A = 0.30$; brackets denote range for $A \in [0.15, 0.45]$ and R_p uncertainty. ^dAntipodal coordinate of host star. β_{ecI} is ecliptic latitude. ^eContinuous 90-day tracking Δv_{90} ; uncertainty reflects orbital eccentricity modulation. ^fRadial-velocity candidate; parameters from Anglada-Escudé et al. (2016); Bonfils et al. (2018); Dreizler et al. (2020, 2024); Astudillo-Defru et al. (2017); Feng et al. (2017).

locked ($P_{\text{rot}} = P_{\text{orb}}$), a precise orbital period does not by itself guarantee synchronous rotation. Non-zero orbital eccentricity drives pseudosynchronization to higher spin rates ($P_{\text{rot}} < P_{\text{orb}}$), capture into spin–orbit resonances (e.g., 3:2), and longitudinal librations, while atmospheric thermal tides can maintain asynchronous spin. Characterizing the true rotation period to $\sim 10^{-4}$ via precursor photometric light curves remains essential.

6.2 Deep-space transportation and swarm sizing

Cruising to $z = 650$ AU within 20 yr requires $v_\infty \simeq 154 \text{ km s}^{-1}$ (32.5 AU yr^{-1}), far exceeding chemical propulsion ($3\text{--}4 \text{ AU yr}^{-1}$; Turyshv 2026d). Deep-perihelion solar sailcraft deploying near $r_p = 0.05 \text{ AU}$ ($10.7 R_\odot$) achieve $v_\infty \simeq 105\text{--}155 \text{ km s}^{-1}$ at system areal densities $\sigma_{\text{tot}} \simeq 2.3\text{--}$

4.9 g m^{-2} . A 100 kg craft requires a $140 \times 140 \text{ m}$ sail operating near $\sim 740 \text{ K}$ (Helvajian et al. 2023; Turyshv 2026d).

Beyond 500 AU, solar flux is $> 4 \times 10^5$ times weaker than at 1 AU; all operational power must be nuclear. The Atomic Planar Power for Lightweight Exploration (APPLE) architecture—modular $10 \times 10 \times 1.7 \text{ cm}$ tiles integrating ^{238}Pu with thermoelectrics and solid-state batteries delivering $\geq 1.2 \text{ W}_e$ and 5 Wh storage per tile (NASA/Aerospace Corp. 2022)—scales to $\sim 10^2 \text{ W}_e$ at masses compatible with sailcraft.

Photometric science data volume is modest: 65,536 samples at 32 bytes total $\sim 2 \text{ MB}$ (or $\sim 2 \text{ GB}$ for megapixel maps). Total mission telemetry (navigation, star trackers, health, inter-craft ranging) will dominate the downlink, but laser relays hopping along the string-of-pearls swarm

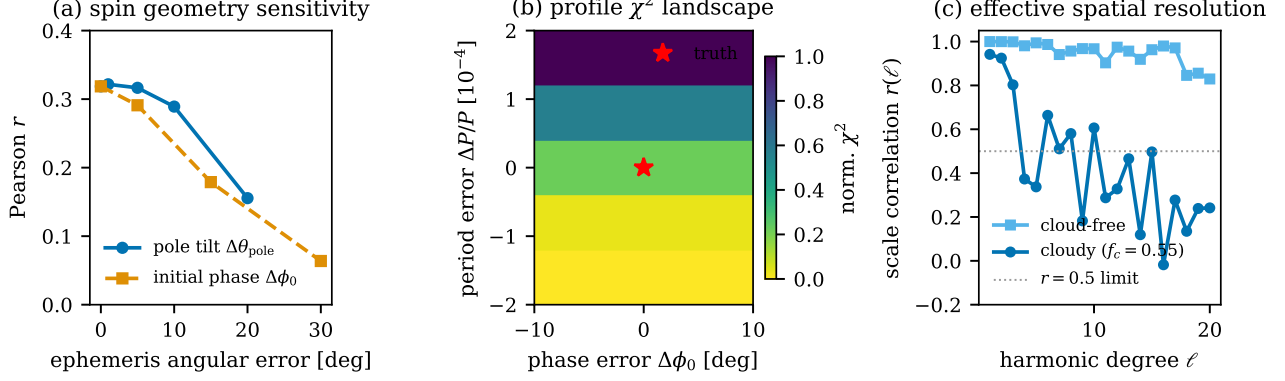


Figure 6: Extended robustness suite. (a) Sensitivity to spin pole orientation tilt $\Delta\theta_{\text{pole}}$ and initial phase $\Delta\phi_0$. (b) Profile likelihood χ^2 landscape over period error and phase offset, showing a sharp global minimum at truth (0,0). (c) Scale-dependent correlation $r(\ell)$ resolving spherical harmonic features to $\ell_{\text{eff}} \approx 20$ (cloud-free) and $\ell_{\text{eff}} \approx 4\text{--}5$ (cloudy).

easily support tens to hundreds of kb s^{-1} (Helvajian et al. 2023). weather.

Re-evaluating swarm extrapolations: Our cadence law proves that revisits improve fidelity by sampling independent cloud states. However, extrapolating the measured trend through 64 revisits to suggest that “several hundred visits yield $r \geq 0.8$ ” must be qualified as an asymptotic architectural scenario rather than a proven requirement. In reality, slew/settle overheads, non-Lambertian scattering, correlated coronal systematics, and spatial cloud correlations impose diminishing returns. Furthermore, the distinction between registered visits per pixel and number of physical spacecraft must remain explicit: 64 registered visits can be delivered by 16 spacecraft operating over 4 successive 90-day campaigns, or by a 64-craft swarm in a single 90-day window.

7 Discussion and Limitations

Several simplifications bound this study: (i) We adopt the monopole, aperture-averaged SGL response; the solar quadrupole ($J_2 = 2 \times 10^{-7}$) creates astigmatic caustics that demand a J_2 -dependent deconvolution library (Turyshv & Toth 2021). (ii) Solar coronal background is modeled as Gaussian shot noise with mean subtracted; unmodeled coronal streamers, low-rank drifts, and instrumental flat-field systematics require sub-ppm calibration (Turyshv 2026b). (iii) Clouds are modeled as 2D single-layer screens without 3D vertical structure, scattering phase functions, or specular ocean glint. (iv) Image-plane navigation is assumed known; uncompensated pointing jitter and ephemeris drift will act as an additional blur kernel.

These limitations do not diminish our core findings: that rotation-aware inversion eliminates temporal aliasing penalties under characterized geometry, and that observing cadence is the primary design lever against planetary

8 Conclusions

1. Regularized generalized least squares on time-tagged samples (TDI) recovers a rotating, cloud-free exoplanet with $r = 0.990 \pm 0.001$ (SSIM = 0.918 ± 0.002), demonstrating that planetary rotation is tractable under accurately characterized spin and viewing geometry.

2. Clouds dominate the error budget, exceeding coronal shot noise by factors of 14–23 per sample. At fixed single-campaign resources, fidelity collapses to $r \simeq 0.26$ under Earth-like clouds.

3. Factorial ablation isolates the causal driver of the cadence law: frequent revisits improve correlation ($r = 0.14 \rightarrow 0.54$) because revisits sample independent weather states (r_{iid} tracks r_{inst}), while exposure smear and conditioning are secondary. Slew overheads set a practical duty cycle limit ($\eta \simeq 91\%$ at 450 s).

4. Component-wise ablation proves slot deflation supplies the primary gain ($\Delta r = +0.041$, $p < 10^{-3}$), removing disk-integrated cloud modes.

5. Precursor ephemerides require spin pole tilt to $\lesssim 5^\circ$, initial phase to $\lesssim 5^\circ$, and period to $\sim 10^{-4}$, refinable in-flight via profile likelihood. Scale correlation resolves features to $\ell_{\text{eff}} \approx 20$ (cloud-free) and $\ell_{\text{eff}} \approx 4\text{--}5$ (cloudy), detecting the land–ocean dichotomy at $p < 0.02$.

6. Physical uncertainties are propagated for the five nearest confirmed targets: Ross 128 b offers the best compromise with an in-ecliptic focal line; candidate τ Cet e would provide 50-times gentler image-plane dynamics. Multi-spacecraft swarms are motivated as an asymptotic architectural path to mitigate weather noise.

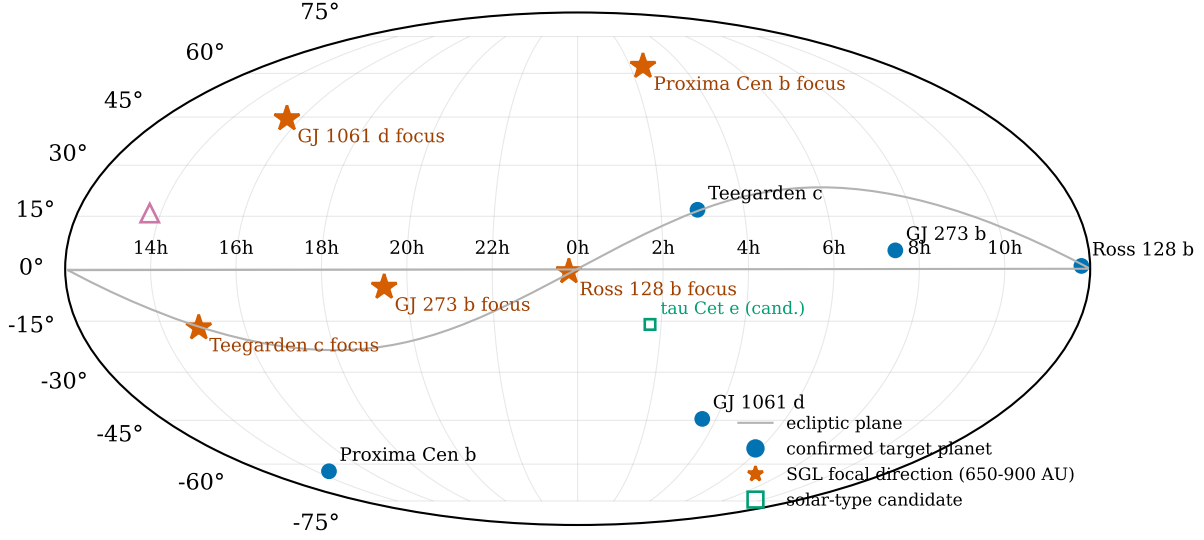


Figure 7: Celestial placement (equatorial Mollweide) of the five confirmed targets (blue circles), candidate τ Cet e (green square), and their SGL focal directions (orange stars, purple triangle). The gray line marks the ecliptic plane. Ross 128 and Teegarden c focal lines lie in the ecliptic ($\beta_{\text{ecl}} \approx 0^\circ$), minimizing departure $\Delta\nu$.

Acknowledgments

This work used open-source software (NumPy, SciPy, Matplotlib, scikit-image). The author thanks the editor and referees for constructive reviews that substantially strengthened this analysis. No external funding was received.

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